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CS466: Midterm Exam

$2^{64}(T_{DES} + O(1))$

Problem 0. Sign your name below to indicate you followed the UMass Amherst academic honor code in preparing for and taking this exam. Without your signature, your exam will not be graded.

$DES_{K_2}^{-1}(C) = M \oplus K_1$

Problem 1. (50 points.) Define $F\colon \{0,1\}^{120} \times \{0,1\}^{64} \to \{0,1\}^{64}$ as follows for any $K_2 \in \{0,1\}^{56}$ and $K_1, M \in \{0,1\}^{64}$:

M, C

Algorithm $F_{K_1 \| K_2}(M)$:
$W \leftarrow K_1 \oplus M$
$C \leftarrow \mathsf{DES}_{K_2}(W)$
Return C

For i : 0 – 2^{64}
For j : 0 – 2^{56}

(Part A - 10 points.) Prove that F is a blockcipher.

(Part B - 10 points.) What is the running-time of a 3-query exhaustive key search adversary against F?

3-query $+ 2^{112}(T_{DES} + O(1))$

(Part C - 30 points.) Give the most efficient 3-query consistent-key recovery adversary that you can having advantage 1 against F. Give your adversary in concise pseudocode and state and prove your adversary's advantage and resource usage. Your adversary should have running-time proportional to about 2^{56} cryptographic operations.

CKR

(Part D - 10 points extra credit.) Would you expect that your adversary in Part C recovers the target key? Justify your answer.

Part A ① $W = DES_{K_2}^{-1}(C)$ ② $M = W \oplus K_1$

$F_{K_1 \| K_2}^{-1}(C) = DES_{K_2}^{-1}(C) \oplus K_1 = M$

$F_{K_1 \| K_2}(M) \Rightarrow$ ① $DES_{K_2}^{-1}(C) \oplus \cancel{K_1 \oplus K_1} = W$

② $DES_{K_2}(DES_K^{-1}(C)) = C$

F is a blockcipher

$2^{56}(3T_{DES} + O(n))$

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$\boxed{(M, C)}$

This page is for additional work on Problem 1.

(part c)

① Adversary AF

$(M_1, C_1), (M_2, C_2), (M_3, C_3) \leftarrow$ 3-query of E_n

$K_2 \xleftarrow{\$} \{0,1\}^{56}$

$W_1' \leftarrow DES^{-1}(K_2, C_1)$

$K_1 \leftarrow W_1' \oplus M_1$

$M_2^2 \leftarrow DES^{-1}(K_2, C_2) \oplus K_1$

$M_3^3 \leftarrow DES^{-1}(K_2, C_3) \oplus K_1$

If $(M_2^2 = M_2)$ and $(M_3^3 = M_3)$

return $K_1 \| K_2$, else repeated.

② $Adv_F^{CKR}(A) = 1$

$Adv_F^{CKR}(A) = Pr[K' \text{ is consistent with } K]$

Proof: When we use (M_1, C_1) get key K': $K_1 \| K_2$.
If we check it is correct on (M_2, C_2).
(M_3, C_3), which shows that the key is consistent.

③ Resource usage:

3-query + $2^{56}(3T_{DES} + O(n))$

$a = \bar{b}$

$b = c$

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$k \quad l$

Problem 2. (30 points.) Define $G\colon \{0,1\}^{128} \times \{0,1\}^{192} \to \{0,1\}^{128}$ as follows for any $K \in \{0,1\}^{128}$ and $M \in \{0,1\}^{192}$:

Algorithm $G_K(M)$:
Parse M as $a\|b\|c$ where $|a| = |b| = |c| = 64$
$w_1 \leftarrow \mathsf{AES}_K(a\|b)$; $w_2 \leftarrow \mathsf{AES}_K(\bar{b}\|c)$
$y \leftarrow \mathsf{AES}_K(\underline{w_1 \oplus w_2})$ $\quad 0^{64}$ $\quad \boxed{\mathsf{AES}_K(0^{64})}$
Return y

Show that G is not a secure PRF. Give your adversary in concise pseudocode and state and prove your adversary's advantage and resource usage. Generic attacks such as birthday attacks and exhaustive key-search will not earn any points.

$0^{64}, 1^{64}, 1^{64} \Rightarrow \underline{\mathsf{AES}_K(0^{64}\|1^{64})} \oplus \underline{\mathsf{AES}(0^{64}\|1^{64})}$ $\quad 0^{64}$

① Adversary A^{F_n}

$a \xleftarrow{\$} \{0,1\}^{64}$, $a_1 \xleftarrow{\$} \{0,1\}^{64}$

$b \leftarrow \bar{a}$, $b_1 \leftarrow \bar{a}_1$

$c \leftarrow b$, $c_1 \leftarrow b_1$

$M \leftarrow a\|b\|c$, $M_1 \leftarrow a_1\|b_1\|c_1$

$y \leftarrow F_n(M)$, $y_1 \leftarrow F_n(M_1)$

return 1 if $y \oplus y_1 == 0^{64}$ else 0

② $\mathsf{Adv}_F^{PRF}(A) = 1 - 2^{-128}$

$\mathsf{Adv}_F^{PRF}(A) = \Pr[\mathsf{Real}_F \Rightarrow 1] - \Pr[\mathsf{Rand}_F \Rightarrow 1]$

$\boxed{\bar{b} = \bar{\bar{a}} = a}$

1. $\Pr[\mathsf{Real}_F \Rightarrow 1] = 1$

Proof: $a = \text{Random } \{0,1\}^{64}$, $b = \bar{a}$, $c = b \Rightarrow M = a\|b\|c$

$a_1 = \text{Random } \{0,1\}^{64}$, $b_1 = \bar{a}_1$, $c_1 = b_1 \Rightarrow M_1 = a_1\|b_1\|c_1$

<1> $G_K(M) = \mathsf{AES}_K(\mathsf{AES}_K(a\|b) \oplus \mathsf{AES}_K(\bar{b}\|c))$
$= \mathsf{AES}_K(\mathsf{AES}_K(a\|b) \oplus \mathsf{AES}_K(a\|b))$
$= \mathsf{AES}_K(0^{64})$

<2> $C_K(M_1) = \mathsf{AES}_K(\mathsf{AES}_K(a_1\|b_1) \oplus \mathsf{AES}_K(\bar{b}_1\|c))$
$= \mathsf{AES}_K(\mathsf{AES}_K(a_1\|b_1) \oplus \mathsf{AES}_K(a_1\|b_1))$
$= \mathsf{AES}_K(0^{64})$

This page is for additional work on Problem 2.

$$G_K(M) \oplus G_K(M_1)$$
$$= AES_K(0^{64}) \oplus AES_K(0^{64})$$
$$= 0^{64}$$

2. $\Pr[Rand_F \Rightarrow 1] = 2^{-128}$

$Rand_F(M) = C$, $Rand_F(M_1) = C_1$

$\Pr[C == C_1] = 2^{-128}$.

③ Resource usage:

2-query + $2F_n + O(n)$